Every element of an ordinal is an ordinal

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openlogic:set-theory/ordinals:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own lemma environment ordmemberord (source lines 416-418, the \ollabel on line 416); the proof environment begins two lines below it and is its own proof, closing at source line 431.

Editorial note. The proof invokes the author's proposition wo:strictorder by its source label \olref[sth][ordinals][wo]{wo:strictorder} (source lines 429-430). That proposition lives in the well-orderings section near the end of the same frozen file, so its statement and its own proof (source lines 1250-1263) are retained as context: the reference then resolves inside the assembled package and no external source pointer has to be declared.

Editorial note. The context is placed before the counted statement in the assembled body, so the retained context proposition prints with the numbering of the retained front matter and the counted lemma keeps its own.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained items print with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's set-theory/ordinals unit, counted once: every element of an ordinal is an ordinal.

Allowed standard prerequisites (editorial).

- Ordinal. An ordinal is a set that is transitive and well-ordered by membership.. The proof uses transitivity of alpha at source lines 421-422 and 425-427, and well-orderedness of alpha where it invokes the retained proposition (source line 428-430).
- Transitive set. A set b is transitive when c in b implies c subseteq b.. The second half of the proof proves exactly that b is transitive, starting from x in c in b (source lines 425-431).
- Membership is a strict well-order on an ordinal. On an ordinal alpha the membership relation is irreflexive, asymmetric and transitive, and every non-empty subset of alpha has a unique least element (the retained context proposition wo:strictorder).. The transitivity clause is applied to x, c, b in alpha at source lines 427-430.

Proposition 0.1. *If $<$ well-orders A, then every non-empty subset of A has a unique $<$-least member, and $<$ is irreflexive, asymmetric and transitive.*

Proof. If X is a non-empty subset of A, it has a $<$-minimal !!element m, i.e., $(\forall z \in X) z \not< m$. Since $<$ is connected, $(\forall z \in X) m \leq z$. So m is the $<$-least !!element of X.

For irreflexivity, fix $a \in A$; the $<$-least !!element of $\{a\}$ is a, so $a \not< a$. For transitivity, if $a < b < c$, then since $\{a, b, c\}$ has a $<$-least !!element, $a < c$. Asymmetry follows from irreflexivity and transitivity $\square$

Lemma 0.2. *Every !!element of an ordinal is an ordinal.*

Proof. Let α be an ordinal with $b \in \alpha$. Since α is transitive, $b \subseteq \alpha$. So $\in$ well-orders b as $\in$ well-orders α.

To see that b is transitive, suppose $x \in c \in b$. So $c \in \alpha$ as $b \subseteq \alpha$. Again, as α is transitive, $c \subseteq \alpha$, so that $x \in \alpha$. So $x, c, b \in \alpha$. But $\in$ well-orders α, so that $\in$ is a transitive relation on α by 0.1. So since $x \in c \in b$, we have $x \in b$. Generalising, $c \subseteq b$ $\square$